

Statistical model to derive revenue projections for a GB-France electricity interconnector – Equity Investor POV

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1. The Problem

Ofgem's Cap and Floor regime sets a revenue floor below which consumers top up the developer, and a revenue cap above which developers return money to consumers. For, a 2 GW GB-France interconnector, these levels are determined by applying Ofgem's Window 3 financial parameters (published December 2024).

Core gap: No probabilistic estimate of floor and cap breach frequency exists for any Window 3 interconnector. This dissertation fills that gap for IC using five years of post-Brexit Bloomberg hourly price data and a calibrated stochastic model.

2. Research Question

Using a stochastic model of GB-France spread revenues, would a Cap and Floor regime make such a project investible from an equity investor's?

3. Why the Spread Is the Right Focus

An IC earns congestion rent by flowing electricity from the cheaper market to the more expensive one and capturing the price difference. Annual revenue is approximately: capacity $\times$ availability $\times$ average hours of positive spread $\times$ average spread magnitude. Capacity (2 GW) and availability (assumed 95%) are fixed parameters. The regime floor and cap are fixed at Final Investment Decision. The GB-France hourly price spread is the only stochastic variable. Modelling it well is the entire methodological problem.

The post-Brexit GB-France spread exhibits three statistical properties that any adequate model must capture:

- **Mean reversion:** Extreme spreads compress over hours and days as markets re-equilibrate. Confirmed by Augmented Dickey-Fuller tests on post-Brexit data: the spread is stationary and mean-reverting, not a random walk. This is the empirical basis for the Ornstein-Uhlenbeck process used in the spread model.
- **Volatility clustering:** Periods of high spread volatility follow periods of high volatility. Sapio and Spagnolo (2020) document GARCH persistence coefficients of 0.93 in comparable electricity markets, confirming that time-varying volatility is a structural feature.
- **Jump behaviour:** Discrete price spikes from supply or demand shocks (unexpected nuclear outages, the 2021-22 gas crisis) that smooth diffusion models miss entirely. Cartea and González-Pedraz (2012) demonstrate that jump components improve in-sample model fit by 20-48% on European interconnector spread data.

Note on transferable risks: This analysis focuses on quantifiable market price risk, which is the principal driver of floor/cap breach probability. Regulatory risk (changes to cap and floor parameters, policy

reversal), construction risk (delay to Final Investment Decision), and interconnection availability risk are not modelled stochastically.

4. Sources used

The methodology adapts two peer-reviewed frameworks to the specific regulatory and market context of IC's Cap and Floor assessment. Each step below identifies the source paper, what it contributes, and how it is applied here.

4.1. Stochastic model of electricity prices

Adapted from: *Abadie, L.M. and Chamorro, J.M. (2021) 'Evaluation of a cross-border electricity interconnection: The case of Spain-France', Energy, 233, p. 121177.*

The methodology used by Abadie and Chamorro (2021) evaluates the economic value of a cross-border interconnector by combining a continuous-time stochastic price simulation with a capacity-constrained econometric flow model.

- 4.1.1. **Independent Domestic Price Modeling:** Instead of modeling the price gap between the two countries directly, the authors model the spot electricity prices of Spain and France separately. Each country's price is split into a deterministic component (which accounts for hourly and daily seasonality, time trends, and non-working days) and a stochastic component.
- 4.1.2. **Correlated Jump-Diffusion (OU) Equations:** The stochastic component for each country is modeled using an Ornstein-Uhlenbeck (OU) mean-reverting process with jumps. Unlike other models, the jump sizes are assumed to be normally distributed. To account for shared market shocks, the stochastic equations for Spain and France are mathematically linked through a correlation coefficient.
- 4.1.3. **Monte Carlo Simulation:** Using an Euler discretization method, the authors run 10,000 Monte Carlo simulations to forecast individual daily and hourly prices for both countries over a future horizon (2020-2022). The simulated hourly price gap is then derived simply by subtracting the simulated French price from the Spanish price.
- 4.1.4. **Tobit Model for Power Flows:** Because the physical interconnector has strict limits, historical power flows are "censored" (meaning they cannot drop below 0 MWh or exceed the 3,500 MW maximum capacity). To accurately capture the relationship between historical price gaps and power flows without skewing the data, the authors use Maximum Likelihood Estimation (MLE) to fit a Tobit regression model instead of standard linear regression.
- 4.1.5. **Revenue Distribution Calculation:** Finally, the authors feed the simulated hourly price gaps (from Step 3) into the Tobit model equations (from Step 4) to predict realistic, capacity-bound hourly export and import volumes. By multiplying these simulated volumes by the price gaps and subtracting a transmission cost (e.g., €5/MWh), they generate a probability distribution of the interconnector's expected transmission income and congestion rents.

4.2. Jump Diffusion Model

Adapted from: *Cartea, A. and González-Pedraz, C. (2012). 'How much should we pay for interconnecting electricity markets? A real options approach'. Energy Economics, 34(1), pp.14–30.*

Cartea and González-Pedraz (2012) developed a three-component model for electricity price spreads between interconnected markets, calibrated to European day-ahead data. Their paper was written to

value interconnector leases as real options; this dissertation borrows the statistical model of spread behaviour only, not the option pricing machinery. The model decomposes the spread into:

- **A deterministic seasonal component $f(t)$:** estimated by OLS regression of the Bloomberg spread on monthly dummy variables and a linear time trend. This captures predictable seasonal patterns (winter peaks, summer troughs) and the secular drift in the GB-France differential.
- **A mean-reverting Ornstein-Uhlenbeck process $X(t)$:** capturing continuous, smooth fluctuations around the seasonal mean. Estimated by AR(1) regression on the jump-cleaned detrended spread. The regression coefficient gives the mean-reversion speed.
- **A compound Poisson jump process $Y(t)$:** capturing discrete shocks from events such as nuclear outages or sudden demand surges. Positive and negative jumps are estimated separately, with arrival rates fitted by season and jump sizes fitted to exponential distributions. Jumps are detected using an iterative algorithm, which distinguishes genuine dislocations from rapid mean reversion by testing whether the new price level falls outside the confidence interval of $f(t)$. This distinction matters because misclassifying mean reversion as jumps artificially inflates jump frequency and distorts tail probabilities.

5. Methodology

5.1. Methodology

The forecasting framework employs a two-stage hybrid approach, combining a recursive semi-parametric jump-detection filter with a continuous-time stochastic price simulation and a capacity-constrained econometric flow model. The process involves:

- 5.1.1. **Deterministic Detrending:** Estimating and removing the predictable seasonal and trend components from the daily log spot prices of both target markets (Abadie and Chamorro, 2021).
- 5.1.2. **Semi-Parametric Jump Filtration:** Applying a recursive algorithm to the stochastic residuals to accurately disentangle genuine structural price jumps from instances of fast Ornstein-Uhlenbeck (OU) mean-reversion (Cartea and González-Pedraz, 2012).
- 5.1.3. **Calibrate the OU Process:** Once those extreme jumps are separated, the data we have left represents the "normal," everyday market fluctuations. We use this clean, non-spiky data to calculate the exact parameters for the OU process – specifically, the standard day-to-day volatility (σ) and how fast normal prices naturally pull back to their average (the mean-reversion speed, κ).
- 5.1.4. **Bivariate Monte Carlo Simulation:** Forecasting future daily spot prices using an Euler discretization of correlated OU jump-diffusion processes, calibrated with the cleansed historical data (Abadie and Chamorro, 2021).
- 5.1.5. **Censored Econometric Flow Modeling:** Translating the simulated inter-market price gaps into physical power flows utilizing a Tobit model estimated via Maximum Likelihood Estimation (MLE), strictly enforcing the asset's technical capacity boundaries (Abadie and Chamorro, 2021).

Expected outputs

- P10, P25, P50, P75, P90 RVI

- Floor Breach Probability ie number of paths below floor for that yr / 10,000
- Cap Breach Probability ie number of paths above cap for that yr / 10,000 (RIIO3 Finance Annex - Cost of Equity; Ofgem)
- The Equity Value Corridor Width
[Corridor width = P50 revenue – Floor level] (CEPA, 2025; Spohr et. al, 2024; EIOPA - Only for european investors) {At least 200-375 bps higher than GILTs OR IRR >10.1%; Higher the better}
- Revenue at Risk [Revenue at Risk = P50 revenue – P10 revenue] (Blackstone)
- Upside Capture Ratio
[Theoretical maximum corridor = Cap level – Floor level; Actual equity corridor at P50 = P50 revenue – Floor level; Upside capture ratio = Actual corridor / Theoretical maximum corridor] (Ben Morton)
- Driver attribution [Identify main drivers for spread variance] (Shapley value decomposition)
 - Long-run spread trend
 - Cannibalism from interconnectors
 - Jump events
 - Regime-switching volatility
 - Nuclear new built - Hinkley
 - GB-France correlation
 - Wind CfD acceleration

6. Statistical defence of the methodology

- **Why use Cartea's Filter over Standard Non-Parametric Jump Tests?** Electricity prices exhibit an extreme statistical anomaly: large price spikes followed immediately by massive drop-offs as the market reverts to the mean (Cartea and González-Pedraz, 2012). Standard filters, such as the Lee-Mykland (L-M) test, frequently misidentify these rapid mean-reverting drops as new, downward Poisson jumps, leading to a severely misspecified jump intensity parameter (Cartea and González-Pedraz, 2012). We apply Cartea's filter because it subjects innovations greater than 3 standard deviations to a rigorous secondary test: if the subsequent price corrects back into the 95% confidence interval of the seasonal trend, it is classified correctly as OU mean-reversion, not a jump (Cartea and González-Pedraz, 2012).
- **Why use the Tobit Model instead of Ordinary Least Squares (OLS)?** Applying standard OLS linear regression to interconnector flows violates the core OLS assumption of homoscedastic and normally distributed error terms (Abadie and Chamorro, 2021). Historical flow data accumulates vast probability masses exactly at 0 MW (left-censored) and the maximum capacity limit (right-censored) (Abadie and Chamorro, 2021). OLS regression on censored data yields biased and inconsistent slope estimates (Abadie and Chamorro, 2021). Utilizing MLE via a Tobit model perfectly accounts for the censored distribution tails, generating consistent parameters that ensure simulated export volumes obey strict physical realities (Abadie and Chamorro, 2021)

7. Driver Decomposition

To test how different real-world risks impact the investor's bottom line, we directly modify specific metrics within the equations:

- **Long-run spread trend:** We alter the time-trend parameter (b_5) to widen or narrow the structural price gap between the two countries over time.
- **Jump events (Extreme Weather):** We scale the jump frequency parameter (λ) and the average jump size (μ, σ) to capture windfall profits during scarcity events.
- **Market Correlation:** We adjust the correlation coefficient (ρ) linking the two countries. Higher correlation means prices move together, destroying the spread arbitrage.
- **Nuclear new build (Hinkley):** We force a downward shift in the long-term equilibrium target (a/k) for the UK, permanently lowering its simulated price baseline.
- **Wind CfD acceleration:** To reflect higher hour-to-hour price swings from intermittent weather, we increase the daily volatility (σ) and jump frequency (λ) parameters in the later years of the simulation.
- **Regime-switching volatility:** To simulate an energy crisis, we override the base volatility metric (σ), temporarily multiplying it to reflect months of market stress.
- **Cannibalism from new interconnectors:** Applied as a direct percentage penalty to the final simulated congestion rent to account for market saturation in future years.

5. Data

All data required for the analysis is confirmed available. The table below sets out each input, its source, its role in the model, and its availability status.

Dataset	Source & Series	Role in Model	Status
GB hourly day-ahead prices	Bloomberg Terminal N2EXEH01-24 Jan 2021–Mar 2026	Primary input for spread construction and all model estimation	Downloaded ✓
FR hourly day-ahead prices	Bloomberg Terminal PWNXFR01-24 Jan 2021–Mar 2026	Primary input for spread construction and all model estimation	Downloaded ✓
GBP/EUR exchange rate	Bloomberg Terminal EURGBP Curncy Matching period	Convert French prices to GBP/MWh for spread construction	Downloaded ✓
IC FA and MA revenues	Arup Market Modelling Report, March 2024 (Ofgem consultation portal)	Source of cannibalisation coefficient (FA–MA gap per GW)	Publicly available ✓
Ofgem Window 3 floor and cap levels	Ofgem Decision, December 2024	Floor threshold against which breach probability is calculated	Publicly available ✓
Window 3 connection schedule	ENTSO-E TYNDP 2022	Timing of cannibalisation deflation year-by-year	Publicly available ✓

Key References

- Arup Group (2024). Market Modelling Analysis for Cap and Floor W3 and Offshore Hybrid Assets. March 2024. Available via Ofgem consultation portal.*
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